PERFECT COMPLEXES ARE COMPACT ON QUASI-COMPACT SPACES WITH QC BASIS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Duality for perfect complexes. Cohomology of Sheaves, Lemma cohomology-lemma-dual-perfect-complex. Hom(K,M) = H^0(X,K^ or otimes ^L M).
- Cohomology commutes with sums. Cohomology of Sheaves, Lemma cohomology-lemma-quasi-separated-cohomology-colimit. H^p(X,-) commutes with direct sums under the qc hypotheses.
- Compactness of j_!O_U. Cohomology of Sheaves, Lemma cohomology-lemma-when-jshriek-compact. Part (b) instance.
- Leray acyclicity. Derived Categories, Lemma derived-lemma-leray-acyclicity. RGamma(X,M) = Gamma(X, oplus I_i^*).

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 10645–10645. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01FF: sources/stacks-chow/tags.txt, source lines 1868–1868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0F5Z: sources/stacks-chow/tags.txt, source lines 18954–18954. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 013K: sources/stacks-chow/tags.txt, source lines 1438–1438. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07D9: sources/stacks-chow/tags.txt, source lines 9404–9404. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a ringed space. Assume that the underlying topological space of X has the following properties:*

(1) *X is quasi-compact,*
(2) *there exists a basis of quasi-compact open subsets, and*

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(3) *the intersection of any two quasi-compact opens is quasi-compact.*

Let K be a perfect object of $D(\mathcal{O}_X)$. Then

(a) *K is a compact object of $D^+(\mathcal{O}_X)$ in the following sense: if $M = \bigoplus_{i\in I} M_i$ is bounded below, then $\mathrm{Hom}(K, M) = \bigoplus_{i\in I} \mathrm{Hom}(K, M_i)$.*

(b) *If X has finite cohomological dimension, i.e., if there exists a d such that $H^i(X, \mathcal{F}) = 0$ for $i > d$, then K is a compact object of $D(\mathcal{O}_X)$.*

Proof. Let $K^\vee$ be the dual of K, see Lemma 08DQ. Then we have

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K, M) = H^0(X, K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} M)$$

functorially in M in $D(\mathcal{O}_X)$. Since $K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} -$ commutes with direct sums it suffices to show that $R\Gamma(X, -)$ commutes with the relevant direct sums.

Proof of (b). Since $R\Gamma(X, K) = R\,\mathrm{Hom}(\mathcal{O}_X, K)$ and since $H^p(X, -)$ commutes with direct sums by Lemma 01FF this is a special case of Lemma 0F5Z

Proof of (a). Let $\mathcal{I}_i$, $i \in I$ be a collection of injective $\mathcal{O}_X$-modules. By Lemma 01FF we see that

$$H^p(X, \bigoplus\nolimits_{i\in I} \mathcal{I}_i) = \bigoplus\nolimits_{i\in I} H^p(X, \mathcal{I}_i) = 0$$

for all p. Now if $M = \bigoplus M_i$ is as in (a), then we see that there exists an $a \in \mathbf{Z}$ such that $H^n(M_i) = 0$ for $n < a$. Thus we can choose complexes of injective $\mathcal{O}_X$-modules $\mathcal{I}_i^\bullet$ representing M_i with $\mathcal{I}_i^n = 0$ for $n < a$, see Derived Categories, Lemma 013K. By Injectives, Lemma 07D9 we see that the direct sum complex $\bigoplus \mathcal{I}_i^\bullet$ represents M. By Leray acyclicity (Derived Categories, Lemma 015E) we see that

$$R\Gamma(X, M) = \Gamma(X, \bigoplus \mathcal{I}_i^\bullet) = \bigoplus \Gamma(X, \bigoplus \mathcal{I}_i^\bullet) = \bigoplus R\Gamma(X, M_i)$$

as desired. □